Deep Learning Module 1 - Detailed 5 Marks Answers

Perceptron

A perceptron is the simplest form of an artificial neural network and serves as the fundamental building block of deep learning. It is inspired by the biological neuron of the human brain. A perceptron receives multiple input values, multiplies them by corresponding weights, adds a bias term, and passes the result through an activation function to produce an output. The main purpose of a perceptron is to perform binary classification tasks. During training, the weights are adjusted to minimize classification errors. A perceptron can successfully solve linearly separable problems but cannot solve complex non-linear problems such as the XOR problem. Despite its limitations, the perceptron laid the foundation for modern neural networks and deep learning systems.

Feed Forward Neural Network

A Feed Forward Neural Network (FFNN) is one of the most basic types of artificial neural networks. In this architecture, information flows only in one direction, from the input layer through one or more hidden layers and finally to the output layer. There are no feedback connections or loops. Each neuron processes information received from the previous layer and passes the output to the next layer. During training, the network learns by adjusting its weights to reduce prediction errors. Feed Forward Neural Networks are widely used for classification, prediction, and pattern recognition tasks. They are simple to understand and implement, making them an important foundation for more advanced deep learning architectures.

Activation Functions

Activation functions are mathematical functions used in neural networks to introduce non-linearity into the model. Without activation functions, a neural network would behave like a simple linear model and would not be able to learn complex patterns. Common activation functions include Sigmoid, Tanh, ReLU, and Softmax. The Sigmoid function produces outputs between 0 and 1 and is commonly used for binary classification. The Tanh function outputs values between -1 and 1 and often performs better than Sigmoid. ReLU, or Rectified Linear Unit, is the most widely used activation function because it is computationally efficient and helps reduce the vanishing gradient problem. Softmax is typically used in multi-class classification problems to convert outputs into probabilities. Activation functions play a critical role in the learning capability and performance of neural networks.

Loss Functions

A loss function is a mathematical measure used to determine how far the predicted output of a neural network is from the actual output. The objective of training a neural network is to minimize the value of the loss function. Different types of loss functions are used depending on the problem. Mean Squared Error (MSE) is commonly used in regression tasks and calculates the average squared difference between predicted and actual values. Cross Entropy Loss is widely used in classification tasks and measures the difference between predicted probabilities and actual class labels. A lower loss value indicates better model performance. Loss functions are essential because they guide the learning process and help optimize the network parameters.

Gradient Descent

Gradient Descent is an optimization algorithm used to minimize the loss function in machine learning and deep learning models. It works by calculating the gradient, or slope, of the loss

function with respect to the model parameters and then updating the parameters in the direction that reduces the loss. The process starts with randomly initialized weights and continuously adjusts them through multiple iterations. The learning rate controls the size of each update step. If the learning rate is too large, the algorithm may overshoot the minimum, while a very small learning rate may slow down training. Gradient Descent is one of the most important concepts in deep learning because it enables neural networks to learn from data effectively.

Stochastic Gradient Descent (SGD)

Stochastic Gradient Descent (SGD) is a variation of Gradient Descent in which model parameters are updated using a single training example at a time instead of the entire dataset. This makes the training process faster and more suitable for large datasets. Because updates occur more frequently, SGD often reaches a good solution more quickly than traditional Gradient Descent. However, the updates can be noisy and may fluctuate around the optimal solution. Despite this limitation, SGD is widely used in deep learning because of its efficiency and scalability. Modern optimizers such as Adam and RMSProp are based on concepts derived from SGD.

Backpropagation

Backpropagation is the learning algorithm used to train neural networks. It works by calculating the error between the predicted output and the actual output and then propagating this error backward through the network. During the forward pass, the network generates predictions. The loss function then measures the prediction error. In the backward pass, gradients are computed for each weight using the chain rule of calculus. These gradients indicate how much each weight contributed to the error. The weights are then updated using an optimization algorithm such as Gradient Descent. This process is repeated over many iterations until the network achieves satisfactory performance. Backpropagation is the key mechanism that allows deep neural networks to learn complex patterns from data.

Regularization

Regularization is a technique used to prevent overfitting in machine learning and deep learning models. Overfitting occurs when a model learns the training data too well, including noise and irrelevant patterns, resulting in poor performance on unseen data. Regularization improves the model's ability to generalize. Common regularization methods include L1 Regularization, L2 Regularization, and Dropout. L1 Regularization adds a penalty based on the absolute values of weights, while L2 Regularization adds a penalty based on squared weight values. Dropout randomly deactivates neurons during training to reduce dependency on specific neurons. By controlling model complexity, regularization helps improve prediction accuracy on new data.

Machine Learning vs Deep Learning

Machine Learning and Deep Learning are related fields within Artificial Intelligence. Machine Learning focuses on developing algorithms that learn patterns from data and make predictions. It often requires manual feature extraction and works effectively on smaller datasets. Deep Learning is a subset of Machine Learning that uses artificial neural networks with multiple hidden layers to automatically learn features from raw data. Deep Learning generally requires larger datasets, higher computational power, and longer training times. While Machine Learning is suitable for structured data, Deep Learning excels in handling unstructured data such as images, audio, and text. Deep Learning has achieved remarkable success in fields such as computer vision, speech recognition, and natural language processing.

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification problems. Unlike linear regression, which predicts continuous values, logistic regression predicts probabilities

and class labels. It uses the Sigmoid activation function to map output values between 0 and 1. The predicted probability is then compared with a threshold value to determine the final class. Logistic Regression is commonly used in applications such as spam detection, disease diagnosis, and customer churn prediction. It is easy to implement, computationally efficient, and provides interpretable results. Although it is called regression, it is primarily used as a classification algorithm.